CHAPTER

5

Contacts, Contours and Periodontium

Microbial plaque is undoubtedly the primary etiological agent in periodontal disease. Nevertheless, other factors do contribute to gingival inflammation and subsequent loss of periodontal attachment. The production of plaque is not as important as the retention of plaque over the tooth surfaces. The retention increases manifolds in case of calculus deposits, poor margins of the restorations and even the presence of a dental restoration.

The outer surface of a restoration is of significance from the periodontal aspects. Proper contact, contour, occlusion, marginal adaptation and surface finish are as important to periodontics as to restorative dentistry. These factors influence the course and direction of masticatory forces, the deflection of food bolus and the collection and retention of deposits and/or food debris. Therefore, it is mandatory to have thorough knowledge of the anatomy and morphology of the teeth to be restored and also the techniques leading to restoration of proper cavosurface margins. Thus, all dental restorations should comply with established requirements for periodontal physiology and health with regard to both surface and functional characteristics. The operator should always take care of the health of periodontium during all restorative procedures.

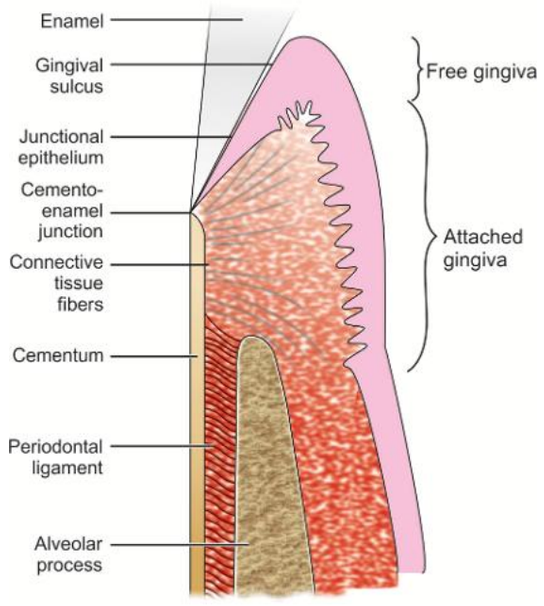
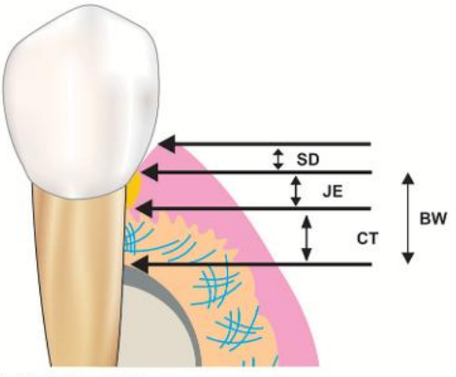


Fig. 5.1: Normal periodontium

DENTOGINGIVAL UNIT

The dentogingival unit and its epithelial and connective tissue covering (Figs 5.1 and 5.2) is described as follows:

a. ***Gingiva:*** It is composed of free gingiva and attached gingiva. The attached gingiva is bound to the cementum and the underlying bone by means of supra-alveolar connective tissue and lamina propria. If restorative procedures are to invade the gingival crevice, approximately 5.0 mm depth of gingiva is involved out of which, 2.0 mm is free gingiva and 3.0 mm is attached gingiva. A second dimension of gingival tissue to be evaluated is 'thickness'. The clinician, should also care for the



SD : Sulcular depth
JE : Junctional epithelium
CT : Connective tissue
BW : Biological width

Fig. 5.2: Dentogingival unit (gingiva, gingival sulcus and biological width)

thickness of the gingiva, which help to tolerate intracrevicular restorative procedures. The clinical test suggests that if a periodontal probe can be seen through the free gingival margin, the ability of that keratinized tissue to support a restoration is doubtful. In case where sufficient thickness of gingival tissue is not present, the margins of the restorations may induce apical migration of the gingiva.

b. ***Gingival crevice:*** The gingival crevice lined with crevicular epithelium, extends from the free gingival margin to the junctional epithelium. Two parameters of gingival crevice are important, one is its depth and the other is its circumference. Normally, the depth of the gingival crevice varies from 2.0–3.0 mm. Depth of the crevice can be excessive or inadequate for restorative dentistry. Excessive crevicular depth is pathognomonic of periodontal disease and restorative procedures should be avoided in such cases. For preparing intracrevicular margins in a tooth, a minimum depth of 1.5–2.0 mm is essential so that the margins are covered by free gingiva. If the crevice depth is less and clinician attempts to place margins of the restoration apical to gingival margin, there is a chance of permanent damage to junctional epithelium and underlying connective tissue. Circumferential aspect of crevice must also be understood. In healthy gingiva, there is hardly any space separating the epithelial lining of crevice from tooth surface. While restoring a tooth with full coverage restoration, care must be taken to avoid distension, which may occur by qualitative or quantitative violation of circumferential aspect of crevice. Quantitative violation is when excessive material is being placed within the crevice, e.g. overhanging margins of the restoration. Qualitative violation implies poor adaptation and roughness of margins of the restoration. Such a violation results in mechanical irritation of crevicular epithelium and also harbours the microbial flora.

c. ***Biological width:*** It is also known as Buffer zone, soft tissue cuff, subcrevicular attachment complex, etc. Biological width is defined as *"combined dimension of the supra-alveolar gingival connective tissue and the junctional epithelium"*. Average measurements for epithelial attachment and connective tissue attachment have been found to be 0.97 mm and 1.7 mm, respectively; thus biological width equals approximately 2.0 mm. A restoration that impinges upon biological width will result in progressive periodontal disease. The margin of restoration should be more than 2.0 mm coronal to alveolar crest.

Encroachment into biological width space is prevalent as the operator attempt to place a margin subgingival rather than intracrevicular. The final position of the proposed gingival margin (dictated by existing restoration, caries or retention features), must be estimated to determine whether crown lengthening procedures are required or not. Biological widths vary according to type of tooth, which progressively increases from anterior to posterior teeth.

Anterior teeth	1.75 mm
Premolars	1.97 mm
Molars	2.08 mm

Restored teeth have wider biological width than non-restored due to longer epithelial attachment around restored teeth. The population variation is depicted in Table 5.1.

PRESERVATION OF PERIODONTIUM

The preservation of healthy periodontium is the most significant factor in long-term prognosis of restored teeth. The health of periodontium depends on the following features.

a. Contour

The buccal and lingual surfaces of teeth possess some degree of convexity and afford protection along with stimulation to the supporting structures during

Table 5.1: Population variation and biological width

85% population	*13% population*	*2% population*
Sulcus—1.0 mm. Normal alveolar crest relation with cementoenamel junction	Alveolar crest and epithelial attachment (EA) are more apical in relatively normal relation, but with long EA	Alveolar crest and epithelial attachment are more coronal in position, but with short epithelial attachment
Least chances of biological width violation, when margins of the restoration are placed subgingivally	Less chances of biological width violation, when margins of the restoration are placed subgingivally	Avoid subgingival margins

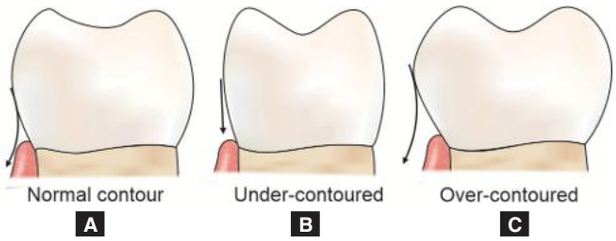


Fig. 5.3: Buccal and lingual curvatures (arrows showing passage of food during mastication): **(A)** Normal contour; **(B)** Under-contoured; **(C)** Over-contoured

mastication. The convexity is generally located at the cervical third of the crown on facial surfaces of all the teeth and lingual surfaces of incisors and canines. The lingual surfaces of posterior teeth have their height of contour in the middle third.

This type of curvature determines the direction in which the food would be pushed cervically during mastication. Normal tooth contours shunt food towards the buccal vestibule and tongue or palate in such a way that the passing food stimulates the intervening tissue rather than irritating them (Fig. 5.3).

Clinical Implications

i. ***Over-contouring:*** If the curvatures are over-contoured, the supporting tissues usually receive inadequate stimulation by passage of food.
Over-contouring of restorations is a much greater hazard to periodontal health than lack of contour since both supragingival and subgingival plaque accumulation may be enhanced by such restorations. Commonly, porcelain fused to metal crowns are over-contoured because of inadequate facial reduction. Such an over-contour interferes with the sealing cuff effect of gingiva against the tooth and the self-cleansing mechanism of gingival sulcus.

ii. ***Under-contouring:*** Under-contoured curvatures may result in trauma to the attachment apparatus. The proximal crown contours are generally flat or concave. This provides adequate embrasure space for the interdental gingiva and allows room for plaque removal. The transitional line angle (the area between the proximal surface and the facial or lingual surface) is also generally flat or concave to form the opening for the embrasure space and house the interdental tissue. The transitional line angle is an important consideration in tooth preparation and full crown restorations. If improperly restored, it will adversely affect the gingiva.

b. Height of Epithelial Attachment

The height of normal gingival tissue, mesially and distally on approximating teeth, is directly dependent upon the heights of epithelial attachment on these teeth. Normal attachment follows the curvature of cementoenamel junction (cervical curvature), if the teeth are in proper alignment and contact. The extent of curvature depends upon:

- Height of contact area above the crown crevice
- Diameter of the crown buccolingually

Crowns of anterior teeth, which are narrower and longer, show the greatest cervical curvature (Fig. 5.4A). Because the crowns of canines distally function as posterior teeth, the cervical curvatures are 1.5–2.0 mm in this region as compared to the mesial curvature that is 2.5 mm. In premolars and molars, the contact level is apical in relation to the total crown length, consequently these teeth do not have high periodontal attachments interproximally (Fig. 5.4B).

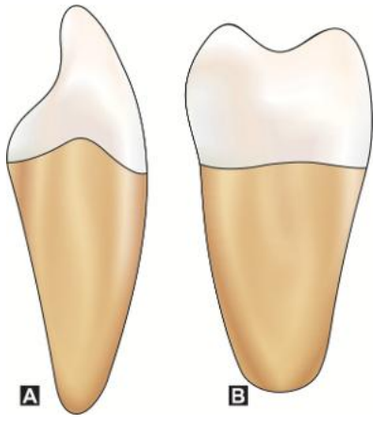


Fig. 5.4: Cervical curvatures: **(A)** Anterior tooth; **(B)** Posterior tooth

Therefore, proximal surfaces of anterior teeth must be carefully prepared for receiving any restoration. The curvatures of posterior teeth are less critical. The height of attachment must be ascertained by careful probing or by continuous observation of landmarks during operation.

c. Proximal Contact Areas

Proximal contact areas are areas on proximal surfaces of tooth crowns where a tooth touches the adjacent tooth when the teeth are in proper alignment.

Proximal contact areas should be considered in two aspects:

- Labial/buccal view showing cervicoincisal or cervico-occlusal location of contact (Fig. 5.5A, B).
- Incisal/occlusal view showing labiolingual location of the contact (Fig. 5.6A, B).

A generalization may be established in locating contact areas faciolingually. Anterior teeth will have their contacts centred labiolingually, whereas posterior teeth will have their contacts slightly buccal

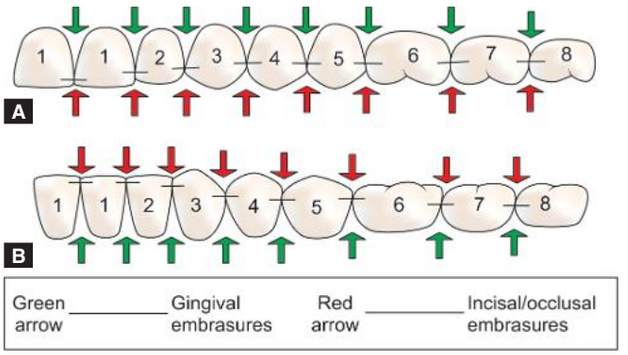


Fig. 5.5: Proximal contact areas (cervico-occlusal view) and gingival and incisal/occlusal embrasures. **(A)** Maxillary teeth; **(B)** Mandibular teeth

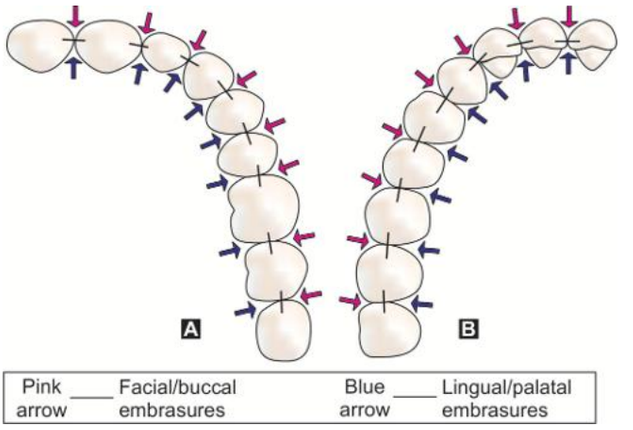


Fig. 5.6: Proximal contact areas (labiolingual view) and facial and lingual/palatal embrasures. **(A)** Maxillary teeth; **(B)** Mandibular teeth

to the center of buccolingual width. The narrower dimension lingually rather than facially causes wide embrasures lingually compared with facial embrasures in all teeth except in maxillary posterior teeth where the vice versa is true.

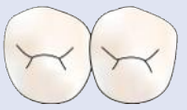
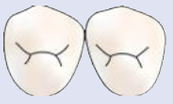
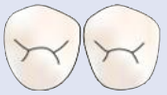
Measuring a Contact Area

Pass the dental floss gingival of the contact from facial to lingual or lingual to facial direction. The two ends are then held parallel in occlusal direction. This measures the faciolingual width of contact. Without removing the floss, the two ends are now held parallel in facial direction. This measures the occlusogingival width of the contact.

In either position, if the parallel strands are more than 1.5–2.0 mm apart, the contact is too broad.

Hazards of faulty reproduction of contact size and location in restorations are summarized in Table 5.2.

Table 5.2: Hazards of faulty contact size and location

Broad contact faciolingually		• Interproximal tissue is over-protected and will not receive proper stimulative massage from excursion of food • Extra breadth of contact prevents food from scouring the embrasures
Broad contact occlusogingivally		• Sticky foods are likely to be held • Encroaches upon gingival tissue • If proximal caries occurs, it will be farther gingivally
Contact too far gingivally		• Creates a wider space between the two proximal surfaces in an occlusal direction. Sticky foods are likely to become packed into this space and eventually force the teeth apart • Impinges upon the interproximal tissue
Contact too far incisal/occlusal		• Frequently observed in amalgam restorations • Crest of gingiva is far from contact and predisposes the proximal surface to caries • Prevents food from being pushed into embrasures
Contact too far buccal/lingual		• Buccal/lingual embrasures are narrow, produce inefficient mastication
Loose contact		• Food lodgement in open contact area develops gingival problems and caries

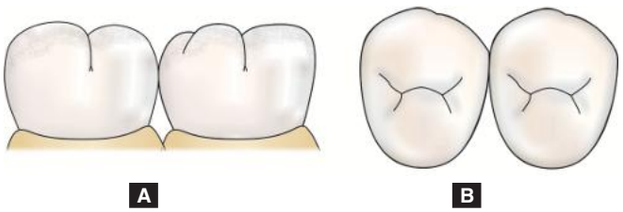


Fig. 5.7: Correct contour. **(A)** Cervico-occlusally; **(B)** Faciolingually

The correct contour cervico-occlusally and facio-lingually is shown in Fig. 5.7A and B, respectively.

A case of poor contact between the restored surfaces of mandibular first and second molars was rehabilitated by giving full crown on mandibular first molar, and half crown on mandibular second molar, creating proper contact and contour of the restoration (Fig. 5.8A, B).

Clinical Implications

The proper proximal contacts facilitate following features:

- Stabilizes the tooth by providing combined anchorage to all the teeth in either arch. The anterior component of force is transmitted through intact proximal contacts. The deflected forces of occlusion through malpositioned/missing contacts may cause displacement of the teeth and create abnormal forces on the periodontium.
- Helps prevent food impaction.
- Protects the interdental papillae by shunting food towards buccal and lingual areas.

d. Embrasures/Spillways

These are V-shaped spaces that originate at the proximal contact area between adjacent teeth and are named according to the direction towards which they radiate (Figs 5.5 and 5.6).

The embrasures, viz. facial, lingual, incisal/occlusal and gingival, are continuous with each other as well as with interproximal spaces.

Clinical Implications

The clinical significances of embrasures are:

- Serve as spillways for escape of food during mastication; therefore, the force being brought to bear on the teeth is also reduced. Food is returned to the occlusal table by movement of the facial tissues and tongue.
- Prevent forcing the food into the contact area. When teeth are attrited and occlusal embrasures are lost, food is easily forced through the contact area.

The size and location of contact area determine the size of embrasure spaces. Too large and too small embrasures provide little protection to the supporting structures.

e. Occlusal Surface/Marginal Ridge

Occlusal surface should be designed as to direct the masticatory forces along the long axis of teeth. Anatomy of occlusal surface should provide well-formed marginal ridges and occlusal spillways to prevent interproximal food impaction.

Marginal ridge is elevated rounded ridge that forms the mesial and distal margins of occlusal surfaces of premolars and molars and mesial and distal margins of lingual surfaces of incisors and canines.

Importance of Marginal Ridge

i. *In general:*
- Maintains balance of teeth in the arch
- Prevents impaction of food interproximally
- Helps in efficient mastication

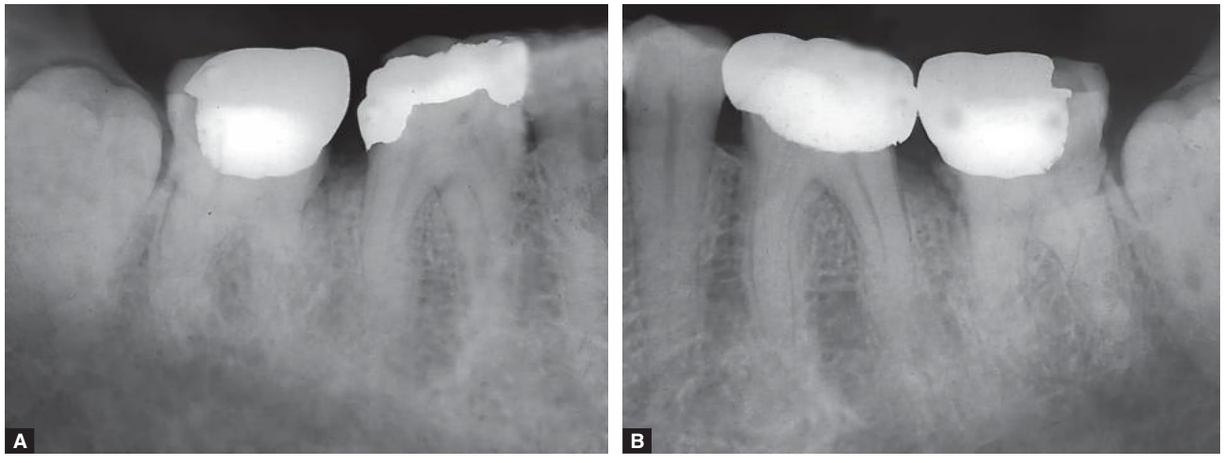


Fig. 5.8: Rehabilitation of contact and contour: **(A)** Preoperative (no contact); **(B)** Postoperative (contact and contour established)

ii. *In restorative dentistry:*
- It is imperative to have a marginal ridge of proper dimensions, i.e. compatible with the dimension of the occlusal cuspal anatomy, creating a pronounced adjacent triangular fossa and producing an adjacent occlusal embrasure.
- A marginal ridge should always be restored in two planes; buccolingually and cervico-occlusally.
- Adjacent marginal ridges should be compatible in height.

f. Margins

The restoration must blend at its margins into the tooth structure without any catch. The explorer is guided from restoration to tooth for analysing the margins. Extra care is mandatory for extra-coronal restorations. The margins should be intracrevicular (confined within the gingival crevice). The margins should not be extended one-half the sulcular depth. Metal restorations seldom provide precision margins. Over-hanging margins lead to interproximal bone loss (Fig. 5.9A, B). Poor margins will invite plaque retention and tend to harbor micro-organisms.

The stepwise rehabilitation procedure of a case in which a single inlay was given in adjacent proximal surfaces of maxillary first and second molars is shown in Fig. 5.10A to I.

g. Surface Finish and Texture

The surface of a restoration should be as smooth as possible to avoid plaque accumulation. Rough surface, especially at subgingival area, results in increased plaque accumulation and subsequently gingival inflammation. It is hypothesized that surface roughness more than 0.2 mm invites bacterial accumulation. Surface free energy is also responsible for bacterial adhesion.

Roughness in the subgingival area is also a major contributing factor in plaque accumulation. Marginal roughness can be because of various factors, viz. scratches at the margins of restoration, luting cement at the margins and inadequate marginal attachment (gap) with the tooth surface.

TRAUMA FROM OCCLUSION

Clinically, the physiologic occlusal pressure is manifested in two forms:
i. Vertical loading
ii. Horizontal loading

i. Vertical Loading

In this form, the applied occlusal force is exerted along the long axis of the tooth. The stress on the periodontal membrane is greater at the apex than along the sides.

ii. Horizontal Loading

In this form, the applied occlusal force is exerted in a direction oblique to the long axis.

In both vertical and horizontal loading, the occlusion pressure on the tooth is balanced by the resistance of the periodontal tissues.

Torques or rotational forces cause both tension and pressure. Under physiologic conditions, they cause bone formation and bone resorption, respectively. However, torques injure the periodontium, if they exceed the adaptive capacity of periodontium.

At rest position, the teeth are apart and no occlusal forces are applied to them. During mastication, the forces applied to the teeth are relatively small. The periodontal ligament fibers are arranged in a definite pattern (most of the fibers are arranged in oblique fashion), which are best suited to tolerate and transmit the vertical forces to the supporting bone. The remaining groups of fibers dissipate the stresses in other directions. Mastication usually does not

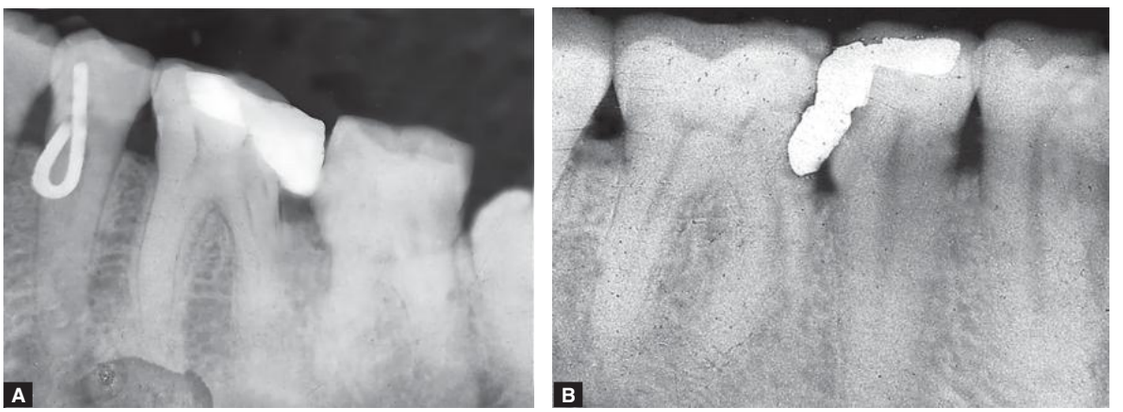


Fig. 5.9: Overhanging margins

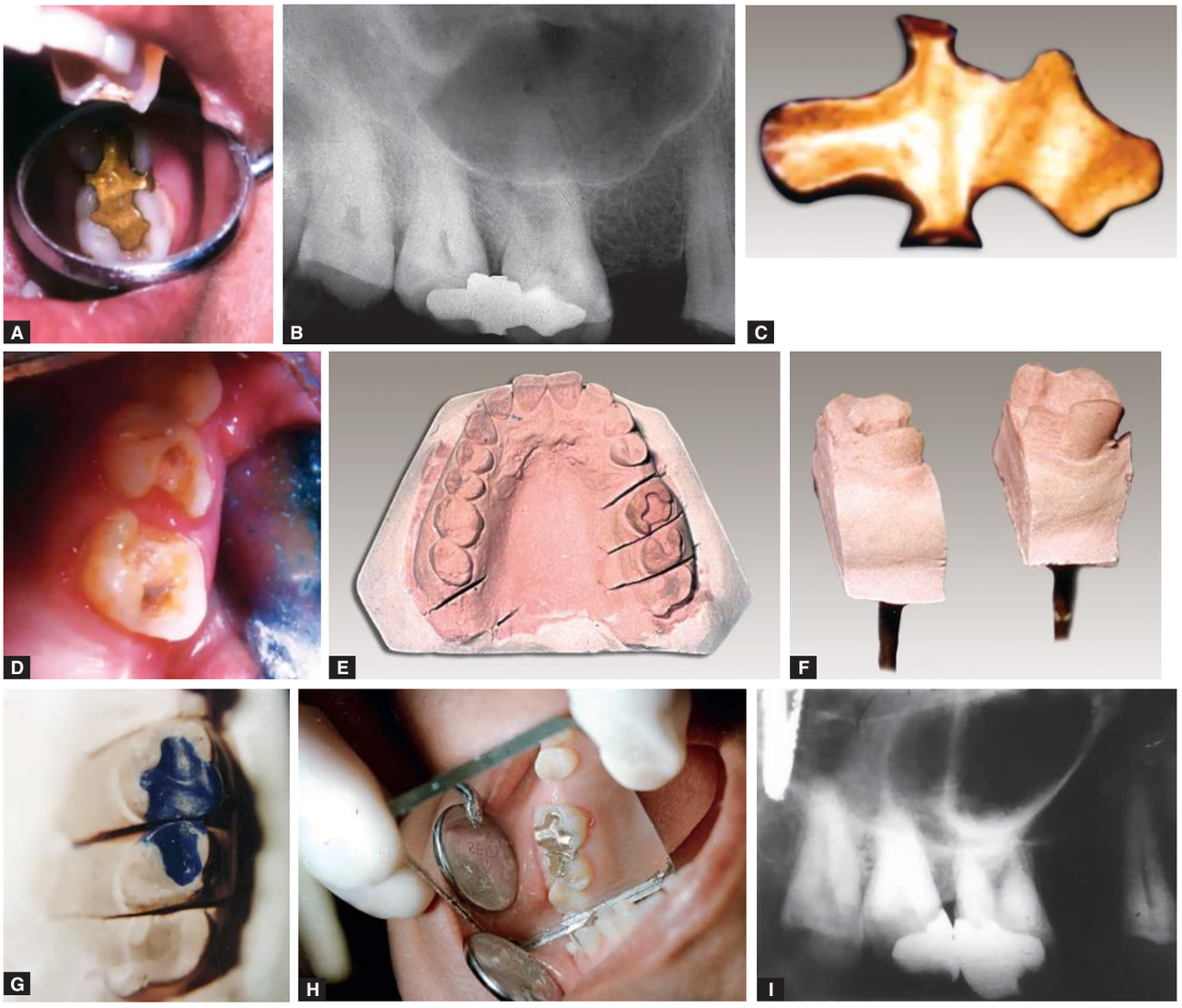


Fig. 5.10: Stepwise rehabilitation procedure of a case in which single inlay was given in adjacent proximal surfaces of maxillary first and second molars. **(A)** Joined inlay in maxillary first and second molars; **(B)** Radiograph showing the joined inlay; **(C)** Inlay removed; **(D)** Modified cavity preparation, with cusp reduced; **(E)** Die showing the preparation; **(F)** Individual dies; **(G)** Pattern making; **(H)** Complete restoration; **(I)** Proper contact and contour as shown in the radiograph

produce trauma as the teeth minimally collide. Trauma usually results when teeth are occluded, other than during chewing. Non-masticatory/parafunctional forces, viz. clenching, bruxism, etc. produce traumatic forces. The forces generated by mastication may have different effects than those generated by habits. Since a habit means frequently repeated action, the occlusal forces are sustained over an extended period of time. Therefore, the parafunctional forces are significant in the etiology of occlusal trauma.

The tooth tries to accommodate the forces of occlusion, which depends upon the adaptive capacity of the periodontium of an individual. However, various factors like magnitude, direction, duration and frequency of force affect the periodontium. When the occlusal forces exceed the adaptive capacity of the periodontium, tissue injury results. This tissue injury is termed as 'trauma from occlusion'. These excessive occlusal forces not only affect the periodontium but also cause painful spasms in masticatory musculature, temporomandibular joint, dysfunction, excessive tooth wear and injury to the pulp tissue.

Etiology

The effect of traumatic occlusion depends on the following factors, however, the effect varies in different teeth and in different individuals:

- The magnitude of force
- The direction of force

- Duration of force
- Frequency of force
- Tissue resistance of the patient

The occlusal relationships, which produce trauma are as follows:

- Uneven occlusal wear (especially when occlusal surface is covered with metal/ceramics).
- Mesial drifting and tipping caused by loss of adjacent teeth.
- Altered clinical crown: root ratio due to periodontal diseases.
- Teeth serving as abutments (clasped teeth, crowned teeth for cantilever bridges, etc.)
- Excessive overbite relationships
- Habit patterns, e.g. bruxism, clenching, tongue thrusting, etc.

According to Glickman, trauma from occlusion may be acute or chronic. Acute trauma from occlusion develops when there is a sudden change in occlusal force, e.g. biting on a hard object, high restorations or prosthetic appliance. These features interfere or alter the direction of occlusal forces on the teeth resulting in tooth pain, sensitivity to percussion and increased tooth mobility. The acute trauma from occlusion subsides by shifting the position of teeth by orthodontic means; or wearing of teeth, or by corrective means, like occlusal adjustment.

Chronic trauma from occlusion is a more frequent finding, which develops due to gradual changes in occlusion, produced by tooth wear, drifting movement, extrusion of teeth. Bruxism, clenching or an acute condition may also develop into a chronic condition.

Trauma from occlusion can be primary or secondary.

a. ***Primary trauma from occlusion:*** When alteration in occlusal forces is the primary etiologic factor in periodontal destruction, e.g. insertion of high filling/prosthesis, drifting/extrusion of teeth, orthodontic movements of teeth into functionally unacceptable positions.

b. ***Secondary trauma from occlusion:*** When adaptive capacity of tissues to withstand normally well-tolerated occlusal forces is impaired by bone loss due to inflammation, e.g. in patients of chronic periodontitis.

Effects of Occlusal Disharmony

The effects of occlusal disharmony can be:

- Discomfort in the region of the temporomandibular joint
- Food impaction
- Abnormal habit
- Pain in and around teeth with improper restorations
- Pulpal disturbances
- Facial pain of obscure origin
- Periodontal abscess
- Cheek biting
- Sensitivity of occlusal/incisal surfaces of teeth due to their abnormal wear
- Mobility of teeth
- Gingival recession
- Pericementitis
- Bruxism
- Unilateral mastication
- Restricted excursion of the mandible resulting in insufficient wear
- Extreme excursions of the mandible resulting in excessive wear
- Migration of upper and lower teeth

Signs of Trauma from Occlusion

The following signs are of diagnostic importance:

Clinical Signs

- Progressive or increased tooth mobility
- Abnormal wear of teeth
- Cheek biting
- Bruxism
- Sharply demarcated linear depressions in the alveolar mucosa, parallel to the long axis of the root and overlying the septal bone

Radiographic Signs

- Widening of periodontal ligament space (Fig. 5.11)
- Thickening of lamina dura (Fig. 5.12)
- Vertical/angular bone loss (Fig. 5.13)
- Rarefaction of bone (Fig. 5.14)
- Condensation of bone (Fig. 5.15)
- Hypercementosis (Fig. 5.16)
- Root resorption (Fig. 5.17)
- Furcation involvement (Fig. 5.18)



Fig. 5.11: Widening of periodontal ligament space

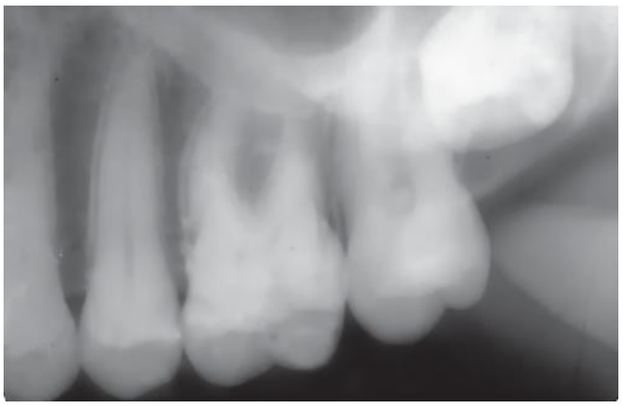


Fig. 5.12: Thickening of lamina dura



Fig. 5.15: Condensation of bone



Fig. 5.13: Angular bone defect



Fig. 5.16: Hypercementosis



Fig. 5.14: Rarefaction of bone



Fig. 5.17: Root resorption

Keeping in view the deleterious effects of trauma from occlusion, the occlusal adjustment is mandatory to create functional relationships favorable with the periodontium. In this, all the opposing teeth make simultaneous contact with their antagonists both in normal and functional relations of the jaw. This can be achieved by various means:

- Reshaping of teeth by grinding
- Restorative measures
- Tooth movement
- Extraction of teeth
- Orthognathic surgery

The objective of occlusal therapy is to achieve therapeutic occlusion that may not conform to the concept of ideal occlusion. The basic objectives of any occlusion therapy are:

- Establishment or maintenance of a stable, reproducible intercuspal position.
- Freedom of movement to and from intercuspal position.



Fig. 5.18: Furcation involvement

- Maintenance of newly established occlusal scheme over a reasonable period of time.
- Establishment of an occlusion with acceptable mastication and esthetics.
- Established occlusion should not interfere with speech.

Occlusal Equilibration

The functional aim of occlusal equilibrium is to improve the following functions:

- Masticatory function
- Temporomandibular function
- Musculature function
- Comfort
- Esthetics
- Phonetics
- Tissue stimulation (relative to the age of the patient)
- Prevention or relief (or both) of trauma
- Oral sanitation
- Elimination of food impaction

Identification of Premature Contacts

Various procedures have been tried to identify and mark the supracontacts or prematurities.

It is advisable to use multiple methods to determine occlusal prematurities. The methods are:

i. ***Study casts:*** Study casts are useful to evaluate the lingual relationships of teeth for better understanding and analysis of all occlusal factors. The areas of food impaction, plunger cusps, abnormal wear, facets and abnormal overbite can be observed in study casts.

ii. ***Radiographs:*** Radiographs are useful adjuncts in diagnosing lesions of the teeth, periodontal space, lamina dura and alveolar bone changes. Bitewing films aid in diagnosing areas with food impaction. Signs of trauma from occlusion include increased width of the periodontal space and thickening of lamina dura; and also vertical/angular bone loss in interdental septum, condensation of the alveolar bone and root resorption.

iii. ***Palpation:*** This method, known as 'fremitus test', is used on upper teeth to determine movement in such teeth when the teeth are in premature centric occlusion. The ball of the index finger is rested lightly against two adjacent upper teeth at a time and the patient is asked to repeatedly open and close in centric occlusion. A tooth in premature centric occlusion will be felt as moving when its antagonist makes contact with it.

iv. ***Visual examination:*** The patient is advised to firmly close the jaw in centric occlusion and the careful observation reveals a slight movement of the tooth in interceptive contact. The gingiva investing the root of a tooth bearing excessive stress may be blanched in some patients.

v. ***Patient's tactile response:*** Patient is asked to slightly open and close the jaw until the first point of occlusal contact is felt. The patient will feel the bouncing on a single portion of a tooth, if there is an interceptive contact.

vi. ***Impression on wax:*** Impression on wax is most reliable to measure prematurities in centric relations. 26 gauge green or pink wax is used. Make wax strips, which can cover occlusal surfaces of teeth. Before making impressions, soften the wax in warm water and place the wax strips over the occlusal surfaces of teeth. Ask the patient to bite on it slowly till the first contact is felt. Take out the impression and chill the wax in ice cold water and examine it against light. Presence of premature contacts is observed as perforation. If there is uniform perforation, then either the wax was too soft, or patient has closed the mouth firmly.

If thick wax is used and the patient is unable to close the mouth, there will be no records over the wax.

These wax records are transferred on models and perforations are marked on study models with pencil.

vii. ***Articulating paper markings:*** Articulating papers are useful for recording interceptive contacts both in the centric and lateral excursions or eccentric relations. In order to differentiate centric relations (CR) from eccentric relations (ER), use two colors of articulating papers, i.e. red and blue; say red for centric relations and blue for eccentric relations. Place the articulating papers on teeth and ask the patient to bite first in centric relations using red articulating paper and then in eccentric relations using blue articulating paper.

Patient is asked to open and close the mouth repeatedly. When prematurity exists, the marking will appear as a lighter area surrounded by a darker zone on the tooth.

The common sources of error are: (i) Smudges may mark more of the tooth surface than actually in contact, and (ii) Some teeth may show markings inadvertently.

Articulating papers are most commonly used to mark and indicate the position of occlusal contacts. The articulating paper is available in different thickness (8 to 200 mm) and in different colors (red, blue, white). The thinner ones are preferred in natural dentition (non-resiliency of teeth); whereas, thicker ones are preferred in denture teeth (resiliency of acrylics). In ceramics and metals, middle thickness is preferred.

The sponge-like nature of the articulating paper stores the color, which is released under pressure. At the area of greater masticatory stresses (heavy contacts), more color is squeezed and produces dark marks; whereas, in light masticatory stresses (light contacts), less color is squeezed, producing less marks. Translucase bonding agent is added to articulating papers and foils to mark on wet surfaces, glazed porcelain or the highly polished metal surfaces.

The difference in foil and paper is depicted in Table 5.3.

The first step is to carry out marking with blue articulating paper (preferably 100 μm thick). The contacts are immediately evident. The second step is to take a thin film (Art foil 8–12 μm), preferably red, to visualize clearly the light spots (Thin areas of the articulating foil marks will clearly differentiate the high spots.)

The smooth fiber reinforced papers (100 μm thick paper impregnated with hydrophilic waxes and pharmaceutical oils) are useful in locating high spots in highly polished metals and ceramics. Articulating silk, made from high quality silk (silk is tear resistant and flexible) adapts perfectly to the anatomy of tooth. Articulating silk is specially suitable for use on laboratory model because one strip can be used for more than once.

The microthin articulating papers (40 mm thick) are suited for two way recording of occlusion, static and dynamic. The first step is to inspect the concentric contact (static occlusion) in blue color followed by second step to inspect the eccentric contact (dynamic occlusion) in red. The color sequence can be changed as per operator's choice. The unique combination of metal foil (shimstock 12) with two-sided color coating enables clear marking on the occlusal surfaces. The centric and eccentric contacts can be shown consecutively with one foil because of the two colors. (Single color coating and without color foils are also available for conventional use.) A white film specially designed for colored modeling waxes is available. (White contact points appear nicely on dark background.) This film also marks effectively on polished metal surfaces. The horseshoe-shaped articulating papers are especially useful for patients who tend to bite unilaterally during the occlusion test due to diminished resilience. Various types of articulation forceps are available for the convenience of the operator. The forceps are also designed for the analysis of proximal contact areas. High-spot indicators are also available for testing the accurate fit of crowns, inlays and also the denture clasps (Arti-spot). Arti-spot is a contact color, which can be applied with a small paint brush. The solvent evaporates leaving a thin film on the surface. The heavy contact destroys the color at that area, visualizing the shining spot (high-spot area). After use, the Air-spot dye can easily be removed with mechanical cleaning and/or alcohol.

A color spray (Air-spray) is available, which is sprayed over the occlusal surfaces of the restoration. A thin film left over the surface, if grinded during occlusal contacts, shows the areas of high contacts. The same can be used for crowns and inlays to note the high-spot areas. The layer left over the restorations can easily be washed off.

Occlusal Adjustments

Occlusal adjustment is a procedure where selective reduction of occlusal areas is carried out to eliminate abnormal occlusal forces and to provide functional

Table 5.3: Difference between paper and foil

Paper	*Foil*
• Size 40 to 200 μm	• Size 8 to 12 μm
• Mark under pressure	• Mark on stroke
• Contact marks are larger	• Contact marks are pinpoint
• Shows different pressure areas	• Visualize high spots precisely
• To check static occlusion	• To check static and dynamic occlusion
• Marks well on wet surfaces	• Equally good in wet conditions

stimulation to the periodontium. The basic idea of occlusal adjustments is to:

i. Eliminate the undesirable occlusal prematurities obstructing the closure of cusps in the fossae, which are responsible for excessive occlusal forces on the periodontium
ii. To create stable mandibular position

Sequence of Occlusal Adjustments

Occlusal correlations should always begin with identification and elimination of the interferences in maximal intercuspal position by the following procedures:

- Shorten extruded teeth
- Reduce plunger cusp
- Correct marginal ridge relationship
- Correct rotated, malpositioned or tilted teeth
- Round sharp cusps
- Create ability to voluntary tap into a constant centric occlusion with patient's head in erect posture.

The armamentarium required for occlusal equilibration is depicted in Fig. 5.19.

Elimination of the slide between centric relation and maximal intercuspal position: Identify the anterior components to the slide and eliminate the interference by correcting the forward facing incline of maxillary teeth and backward facing incline of mandibular teeth.

As a general rule, widen the fossae or embrasure areas at the occluding cusps.

Elimination of protrusive interferences (Fig. 5.20): Interferences during protrusive movements can be eliminated by correcting the distal inclines of maxillary teeth and the mesial inclines of mandibular teeth.

Elimination of retrusive interferences (Fig. 5.21): Interferences during retrusive movements can be eliminated by correcting the mesial inclines of the maxillary teeth and the distal inclines of mandibular teeth.



Fig. 5.19: Armamentarium for occlusal adjustments

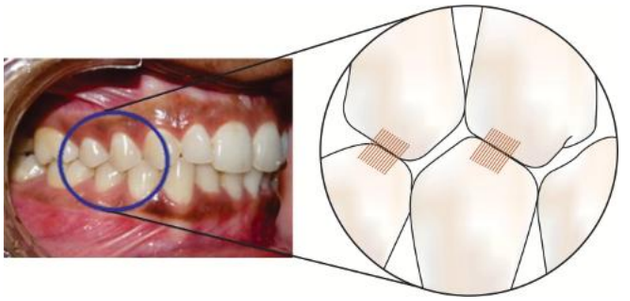


Fig. 5.20: Elimination of protrusive interferences

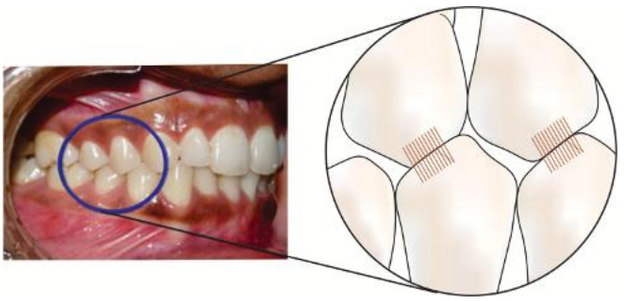


Fig. 5.21: Elimination of retrusive interferences

Elimination of working side prematurity (Fig. 5.22): The working side prematurity is commonly seen in extruded teeth rather than in normal dentition. For such type of interferences, BULL (buccal upper and lingual lower) rule is applied to correct the occlusal prematurities on the working side since these are nonsupporting cusps. The palatal upper and the buccal lower cusps should not be adjusted.

Elimination of non-working side prematurity (Fig. 5.23): The premature contacts pose special problem in the non-working side as they occur on supporting cusps. Their removal, however, is indicated, as they are associated with an increased risk of bone loss, TMJ dysfunctions and mobility. The

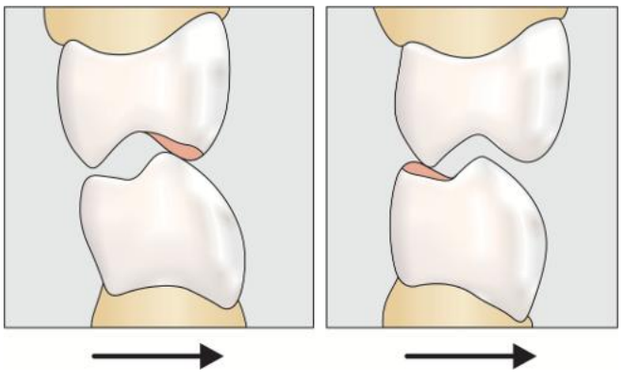


Fig. 5.22: Elimination of working side prematurity

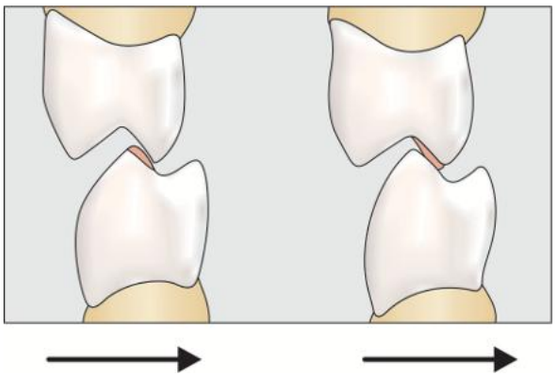


Fig. 5.23: Elimination of non-working side prematurity

palatal slopes of the maxillary buccal cusps and the lingual slopes of the mandibular buccal cusps are modified to eliminate the interferences.

It is not simply a matter of grinding down the premature contacts that will create flattened planes, which will further disrupt the occlusion; it is a planned procedure, which is carried out following the steps as:

i. *Grooving* consists of restoring the depth of the developmental grooves, which have been made shallow by occlusal wear. It is carried out by a tapered cutting tool until a desired depth is achieved.
ii. *Spheroiding* implies reducing the prematurities while restoring the original tooth contour; carried out with a light paint brush stroke, gradually blending the area of prematurity with the adjacent tooth surface. The occlusal height of the cusps is preserved.
iii. *Pointing* implies restoring cups point contours, carried out by reshaping the tooth with cutting instruments.

The occlusion should be analysed periodically; being dynamic in nature, may need minor adjustment every time.

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